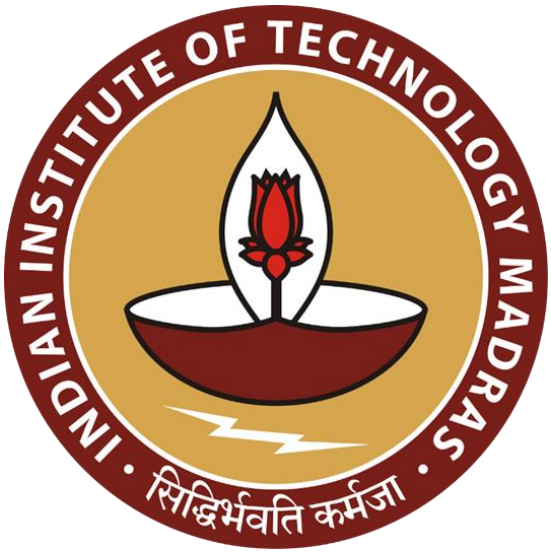
Tracking Student Progress in Software Projects

Milestone 1 Submission for the SE Team Project

Submitted by

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1. Identifying Users of the Application

1.1. Primary Users

These are the direct users who interact with the system regularly to complete essential tasks.

- Instructors – Manage and monitor project progress, set milestones, evaluate submissions, and post announcements.
- Students – Submit project progress, access feedback, view milestones, announcements, and supporting resources.
- TAs – Assist instructors by providing feedback, monitoring student progress, and tracking assigned projects.

1.2. Secondary Users

These users have occasional interactions with the system or provide operational support.

- Admins – Manage user accounts (onboarding instructors and TAs), assign projects, and monitor the overall system for smooth operations
- Institutional IT Staff – Provide technical support, handle system maintenance, and ensure integrations (like GitHub) work correctly.

1.3. Tertiary Users

These users are not directly involved in day-to-day system operations but have an interest in its outcomes.

- Academic Coordinators/Program Managers – Monitor student and course progress at a high level to ensure learning outcomes are achieved.
- External Reviewers or Evaluators – May occasionally access project reports or evaluations for audits, accreditation, or research purposes.

2. User Stories

2.1. For Admin

- **View All Users**

  As an admin,

  I want to view a complete list of users in the system updated in real-time,

  So that I can monitor all registered users and manage their roles efficiently.

- **Onboard New Users**

  As an admin,

  I want to onboard TAs and instructors by approving requests within 10 minutes,

  So that they can start using the platform without delay.

- **Search for a Specific User**

  As an admin,

  I want to search for users by name or role within seconds,

  So that I can quickly locate specific users and manage their permissions.

- **View All Projects**

  As an admin,

  I want to view a complete list of all projects across the system,

  So that I can monitor ongoing work and ensure project compliance.

- **Add New Projects**

  As an admin,

  I want to add new projects into the system and assign instructors and TAs to them within 10 minutes,

  So that instructors can start working on them without delays.

- **Edit a Project**

  As an admin,

  I want to edit project details such as title, deadlines, or assigned instructor and TAs

within 10 minutes,

So that I can ensure all project information stays accurate and up to date.

- **Delete a Project**

  As an admin,

  I want to delete outdated or cancelled projects from the system within 5 minutes,

  So that the project list remains relevant and clutter-free.

- **Search for Projects**

  As an admin,

  I want to search for specific projects by name, description, or instructor within seconds,

  So that I can quickly access relevant project information.

- **Sort Projects**

  As an admin,

  I want to sort projects based on parameters like deadline or progress percentage,

  So that I can prioritize monitoring or action on specific projects.

2.2. For Instructor

- **Milestone Management**

  As an instructor,

  I want to create customizable project milestones for each project within the first week of the semester,

  So that I can track their progress more effectively by comparing their actual progress with the planned milestones at regular intervals (weekly).

- **GitHub Integration for Progress**

  As an instructor,

  I want to automatically track commit histories from GitHub every day,

  So that I can ensure students are making regular progress on their coding tasks and identify inactive teams within 48 hours of inactivity.

- **AI Insights for Progress**

  As an instructor,

  I want to receive AI-based progress predictions weekly,

  So that I can identify students who are likely to fall behind at least two weeks before deadlines and offer them additional support.

- **Analyze Student Documents Using GenAI**

  As an instructor,

  I want to use GenAI to analyze student-submitted documents on demand,

  So that I can quickly gain insights and assess the quality, completeness, and relevance of their work.

- **Evaluate Final Submission and Assign Scores**

  As an instructor,

  I want to evaluate only the final submission of each project and assign scores,

  So that I can focus on the most complete version and provide meaningful feedback.

- **Post Announcements for Project Changes**

  As an instructor,

  I want to post announcements about changes in project deadlines, requirements, or milestones,

  So that students stay informed and can adjust their plans accordingly.

2.3. For TA

- Assist in Project Oversight

  As a TA,

  I want to have access to student progress and milestones,

  So that I can assist instructors in providing feedback and help students stay on track.

- AI Summary for Project Reports

  As a TA,

  I want to receive AI-generated summaries of student reports,

So that I can quickly review and provide comments on progress reports.

2.4. For Student

- **View Upcoming Milestones**

  As a student,

  I want to view my upcoming milestones at least 7 days in advance,

  So that I can prioritize my tasks and complete them on time.

- **View Project Details**

  As a student,

  I want to view the complete details of my projects, including milestones, deadlines, progress, and team members,

  So that I can manage my tasks efficiently and ensure I meet all expectations.

- **GitHub Integration for Submission**

  As a student,

  I want my commits to be automatically pulled from GitHub within 15 minutes of pushing code,

  So that I can have my work updated in the system without manually submitting progress.

- **View Feedback and Receive Notifications**

  As a student,

  I want to receive notifications when my TA provides feedback on my project,

  So that I can review it promptly and make the necessary improvements.

- **View Announcements on Project Changes**

  As a student,

  I want to receive and view announcements about changes in the problem statement, milestones, or deadlines,

  So that I can stay informed and adjust my work accordingly.

- **Access Supporting Documents and Resources**

  As a student,

  I want to view supporting documents and resources related to my project,

  So that I can use them to improve the quality of my work and stay aligned with expectations.